



Tandy Corporation/Radio Shack

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COMPUTER MANUFACTURING DIVISION

Remarks by: Jim Mortensen, Vice President, Computer Manufacturing

Thank you John, thank you ladies and gentlemen. Welcome to Texas, the home of the TRS-80, Computer Manufacturing Division of Tandy Electronics. Our Computer Manufacturing got its start in September of 1977 with its first plant, and appropriately so, the first plant was named Tandy Advanced Products. To respond to customers' demand for Model I, Tandy Advanced Products relocated to larger facilities 3 times, in June of 1978 again in July 1979 and again in August of 1980. Also since that time 5 additional plants have been put into operation to handle the continuing growth of both sales and new products.

The five additional plants are: Tandy Instruments in January 1978. Tandy Business Products in May 1979. Tandy Software Assembly in June 1980 and this year, in the summer of 1982 Tandy Software Assembly will relocate to a new plant, 3 times its current size. Tandy Home Computer in September 1980 and Home Computer will also relocate this year, weather permitting in about 60 days. And our newest plant Tandy Computer Assembly, in June 1981 is located in San Antonio.

With the use of slides, I will take us on a tour of each plant, starting at Tandy Instruments located 7 miles east of downtown Fort Worth.

SLIDE #1

In 1974, Tandy Instruments became plant number 14 in the line up of U.S.A. plants, producing test equipment and calculators for Radio Shack. In January of 78, Instruments became part of the Computer Division.

SLIDE #2

Bringing with it 70,000 square feet of manufacturing space and 400 experienced employees, capable of producing large volumes of highly reliable printed circuit board assemblies.

SLIDE #3

Today, Tandy Instruments' products are produced with the use of computer aided manufacturing machines.

SLIDE #4

Beginning with the first machine, sometimes referred to as an Inventory Preparation tool, we have a computer controlled 100 Station Sequencer, processing 25,000 parts per hour.

SLIDE #5

This machine dispenses in the required order, axial lead components, resistors, capacitors, inductors, diodes, etc.

SLIDE #6

Each component is then tested for value and tolerance, voltage breakdown and proper polarity. Then the parts are re-taped in the proper sequence for the next machine.

SLIDE #7

The components are now loaded onto the computer controlled (VCD), Variable Center Distance, insertion machine. This machine forms the axial lead component into staple shape and inserts them into the P.C.B., then cut and clinch the leads from underneath the board. The machines we use are dual headed and insert two boards simultaneously at a rate of 15,000 components per hour. Currently two such machines are in use at Tandy Instruments.

SLIDE #8

Before we move to the next machine, a little about our Integrated Circuits. Prior to receiving our active components, each I.C. receives an additional battery of tests and additional burn-in time to insure that our boards are loaded with highly reliable devices.

SLIDE #9

Now, with computer controlled (DIP), Dual-inline package, automatic insertion machines, the I.C.'s are selected from one of the 60 holders and with verifier fingers, the I.C.'s are tested, and then inserted, cut and clinched. Three of these machines give us a verifying and insertion rate of 16,500 I.C.'s per hour.

SLIDE #10

The boards, now loaded with components, are moved to the Solder Station. Our P.C. Boards are double sided and high density so for the best solder results, our equipment handles an Organic Acid Flux.

SLIDE #11

The boards are conveyer fed across a foam fluxer and air knives, then over a radiant pre-heater and into an equal flow wave solder generator, driven by dual pumps.

SLIDE #12

The board now receives a hydro cleaning. Again it's conveyer fed through a series of wash and rinse modules that thoroughly emulsify and remove the acid flux and other foreign particles that are water soluble.

SLIDE #13

The last assembly process is a series of work stations installing the remaining components that do not like automation or acid flux solder. The board now receives its final bath in a freon-based pressure and vapor cleaning machine that removes foreign metal particles and Rosin flux.

SLIDE #14

Before leaving Tandy Instruments, each board is evaluated by a series of fault finding test machines to insure there are no component or board traces that are shorted or open.

Now we will leave Tandy Instruments and head for Tandy Business Products and Tandy Home Computers located on the Northeast side, 8 miles from downtown Ft. Worth.

SLIDE #15

Tandy Business Products builds the Model II Business System, Disk Expansion units and the new 8.4 mega-byte Hard Drive.

SLIDE #16

The boards produced at Tandy Instruments arrive here and continue their building and testing process. At this location, with more computer automation, the boards undergo a component-by-component evaluation as power is applied.

SLIDE #17

Next, special test points on the board are evaluated to insure timing and signal levels are within specifications. A Model II is used as the test fixture and it also gives instructions to the operator.

SLIDE #18

Again using a Model II as an instructor and test fixture, alignments are made.

SLIDE #19

All the boards that make up the Model II are plugged into a mother board in a card cage assembly and as an assembly, are burned-in at high temperatures. Then a final acceptance test, and on to the assembly line.

SLIDE #20

On the assembly line, all previously tested and accepted sub-assemblies are brought together and fastened to the main frame as the Model II takes shape.

SLIDE #21

Then on to an elevated temperature burn-in area where it sets for 24 hours.

SLIDE #22

Then to an ambient 24 hour burn-in.

SLIDE #23

Finally, an outgoing audit before being released for shipment.

SLIDE #24

Before leaving Business Products, a quick look at one of the test and exercise rooms for the Hard Drive. This product receives an additional 48 hour elevated temperature burn-in, an extra step to insure reliability prior to releasing for shipment,

SLIDE #25

Next door to Business Products is Tandy Home Computers. Here in the entrance, on display, are the products made in this plant: The Videotex terminal, Color Computer and Modems. Before we go to the next slide, please note on the left side of this display, a picture of shoes. This is one of many types of displays used to communicate at Tandy Home Computer.

SLIDE #26

This plant also requires illustrations, visual aids for all work stations and has instructions, that are written in five languages. This is because the employees are from 17 different countries. The extra effort put into communicating has had a nice side effect on workmanship. Accuracy.

SLIDE #27

An endless assembly line concept is used here and the "one piece" insertion method is utilized, increasing the accuracy of work and increasing productivity over other conventional methods.

SLIDE #28

The final assembly line is also the endless concept. This line is set up so that any of the assemblers may shut down the line should a problem arise. A station light and a revolving red light on the line alerts the supervisor so action may be taken to solve the problem. This type of employee and supervisor involvement has helped Tandy Home Computers produce the highest level of product reliability while being cost effective.

SLIDE #29

Next in line in the burn-in area, a compact shelf/monitor system is used to conserve floor space.

SLIDE #30

Volume is the result of this productive plant - a day's work ready to be shipped out.

We leave here and go to the Northside of Ft. Worth, 7 miles from downtown.

SLIDE #31

Tandy Advanced Products, the first computer plant, got its start building Model I's and continues to support the Model I.

SLIDE #32

Today with 100,000 square feet of production space, Tandy Advanced Products builds in volume, Model III, our desk top business machine, Catalog #26-1066.

SLIDE #33

Disk Drives for the Model III, built in Odessa Texas by Texas Peripherals a joint venture with Tandy Corporation and Datapoint, receive an incoming inspection.

SLIDE #34

Two drives along with the Switch Power Supply and Controller Board are assembled, aligned, exercised and functional tested using a modified Model III as a test fixture.

SLIDE #35

Prior to final assembly, the Central Processing Unit board receives a thorough test and alignment on a point-to-point comparator test fixture.

SLIDE #36

A portion of the day's requirements shown here, is the completed Model III receiving its last 24 hour burn-in.

SLIDE #37

To manage the volume, one thousand and fifty feet of chain move materials in, and computers out of the production area, plus it helps keep the area clean and free from dust, a must in producing computers with floppy disk drives.

SLIDE #38

Before leaving Tandy Advanced Products, at this site we have the future home of Tandy Home Computers - 107,000 square feet.

Now one mile due South is Tandy Software Assembly.

SLIDE #39

At Tandy Software Assembly, all diskettes are duplicated on media technology diskette duplicators and TRS-80 computers. Each diskette receives a surface verification, then a write/read to the diskette and a complete byte-for-byte comparison.

SLIDE #40

Here the diskette and documentation are brought together and then shipped to Tandy Electronics Warehouse and all Radio Shack warehouses world wide.

SLIDE #41

Now we will head south to San Antonio. But before leaving Fort Worth, a quick glance at the future site of Tandy Software Assembly, 135,000 square feet to handle the aggressive growth planned for TRS-80 software.

SLIDE #42

Two hundred eighty-seven miles south of Fort Worth and 4 miles south of downtown San Antonio is Tandy Computer Assembly. The newest addition to the Computer Manufacturing family, producing the increase demands for the Model III.

SLIDE #43

Here the Central Processing Unit board receives a set of 16K RAMS and Level II ROMS, the configuration that makes the Model III a starter system, Catalog #26-1062.

SLIDE #44

Next stop - the Central Processing Unit board burn-in area. Here a space saving rack technique is used, holding a large volume of boards for a 24 hour period.

SLIDE #45

The Central Processing Unit board is now mounted on a RF shield, a keyboard and a power supply, all are added to the cabinet base.

SLIDE #46

The top cabinet houses the Cathode Ray Tube (CRT) and Monitor board. Here a final alignment is made to the video, insuring Raster size, character position and resolution. The top cabinet will now be attached to the base.

SLIDE #47

Here again in volume, the assembly complete, the Model III receives a final 24 hour burn-in. From here these computers will be distributed to all Radio Shack warehouses in the U.S. and Canada.

SLIDE #48

We also have other Tandy plants producing materials to support the Computer Division. Tape duplication located at Tandy Magnetics in Fort Worth, supplies all cassette tape programs.

SLIDE #49

Tandy Wire & Cable also in Fort Worth provides cable harnesses for Model II and III Computers and other wire and cable used by the Computer Division.

SLIDE #50

Leaving Texas for a moment. At Stockton, California, Lika Corporation supplies us and other high technology industries, double sided and multi-layer printed circuit boards.

SLIDE #51

A new Corporation, Matra-Tandy Electronique, S.A. is expected to be formed in the next 30 days. Initial manufacturing will be at Colmar, France where the TRS-80 Model III Microcomputers will be produced and sold to the European market.

SLIDE #52

This ends the tour of Computer Manufacturing Plants. For a final note on growth, here are a couple of numbers our Computer Manufacturing Division has produced. For the first 5 months of this fiscal year as compared to the first 5 months of last fiscal year, sales are up 2.45 - that is times, not %.

Thank you. I hope I have given you all a little insight into what goes on down here in Texas.